

Accidentally writing with permanent marker? - Tracking done by Blackboard

Blackboard^[1] is learning management system that needs no introduction to any Trinity student or staff member - for a lot of us, it's the primary and even sole way to engage with our work, and extremely rarely is an alternative presented or even considered. It provides a single access point for students for almost everything they might need in their daily lives, including course work and assignments, study materials, or even the occasional online video call.

Starting from 2014^[2] Trinity has had ever growing usage of *Blackboard Learn*^[3], the digital platform for learning sold as a service by it's parent company, Anthology. This has culminated in Blackboard becoming a non optional part of student life as it is required for many day to day tasks. In that time, it has gained ever more access to data about students and their lives to a scale many may not realise, which is especially egregious considering there is no feasible way to avoid much of it.

What does Blackboard track?

Blackboard track a vast amount of data from it's users^[4], including:

- **Marketing Information:** *Blackboard* are able to figure out what students may be interested in and share that data with advertising vendors like *Google AdWords* or *Facebook*^[4-1], which is extrapolated from things like a student's module and subject choices or maybe things they have mentioned in submissions which divulge personal data or interests.
- **Personalization Data:** *Blackboard* allow logging in with multiple OpenID Connect (OIDC) providers, which is a standard protocol for determining who a user is based on an external service they have already logged into. Directly, this includes providers such as *Microsoft*, which are then able to explicitly link any user data *Blackboard* may share with them to the logged in user.
- **Student Submissions:** *Blackboard* retains the right to store and potentially share any student submissions^[4-2], such as "responses to quizzes", "files you submit" or "audio" or "video recordings", which may be surprising to many that these submissions can live as long as they do, being directly used for "Analytics and Recommendations", and shared to like *Amazon Web Services* or *Google Analytics*^[4-3].

Additionally, the data from the Blackboard Student Timetable^[5] can trivially be used to predict the location of a student at any given moment during a school term, which presents an obvious and major security risk.

How does Blackboard acquire data?

Blackboard acquires data in a number of different ways depending on how users interact with the platform. First and foremost, the learning management system makes frequent use of various cookie embeddings to track things from login sessions to general identification of who the user is. For the purpose of demonstration, the following were sourced from inspecting the network requests of our very own class page [\[6\]](#). For this experiment it should be noted I am using Librewolf^[7] with uBlock Origin^[8] on NixOS^[9], and your results may vary depending on device, browser and a number of other factors.

In total, a single force refresh (`CONTROL` + `f5`) on most browsers for me caused a download of a grand total of **51** different JavaScript blobs, which is quite high. These can be broken down into a number of sections, including:

- Scripts required to run the blackboard single page application (the majority of the bundle), most of which have been minified with `webpack`^[10] for both code obfuscation and reduced network usage, making them hard to inspect.
- `cookie.js`^[11], the cookie implementation for Blackboard. From this we learn that if we inspect the value of `document.cookie`, we can see some of the cookie values that blackboard stores directly. In my case these were:

Name	Value
<code>COOKIE_CONSENT_ACCEPTED</code>	<code>true</code>
<code>BbClientCalenderTimeZone</code>	<code>Europe/Dublin</code>
<code>JSESSIONID</code>	Arbitrary hashed value
<code>AWSELB</code>	Arbitrary hashed value
<code>AWSELBCORS</code>	Arbitrary hashed value

- `pendo.js`^[12], a known tracking script from *Pendo* used for marketing and user retention purposes, which was blocked automatically by my browser. *Pendo*, in their own words^[13], is a "product intelligence platform" which tracks every interaction with the product. *Blackboard*, being a single page application are using the *Pendo SPA Framework*^[14], which gives a full stream of events identifiable on a per user basis, each of which is configurable by the consumer (*Blackboard* in this case)^[15].
- `js-agent.newrelic.com`^[16], which is from another user tracking platform - *Newrelic*^[17], which seems to focus more on "ads intelligence" and "logs". This runs off of a similar browser agent architecture to *Pendo*^[18]. This seems to be a largely automatically setup browser tracking agent, and as such provides a large number of capabilities which may seem nice for a platform developer but may also be a privacy risk, the most egregious of

which being Full Session Replay ^[19], which allows for replaying up to 4 hours of a given user's exact activity and clicks on a page.

It should be remembered that, as extensive as it seems, this is just a subset of the tracking that can be found through normal methods. Blackboard also state that they collect *Content and Activity*, meaning things like chat logs or submitted materials. Unfortunately, there is no easy way to trace where these end up other than offhand mentions from blackboard of things like *Google AdWords* or *Facebook Business*.

What can you do about this?

While in a lot of cases, there is not much you can do about this tracking short of endlessly petitioning Trinity to switch to some form of open source tracking free alternative to Blackboard, there are steps you can take individually to avoid a lot of the tracking.

For example, your browser setup is crucial to blocking a lot of the tracking. Things like a hardened Firefox setup ^[20] or Librewolf ^[7-1] will block the vast majority of these tracking methods by default, making them good choices for any privacy minded individuals. These typically include options like:

- Automatic clearing of cookies on session exit ^[21]
- Spoofed hardware details (i.e. Librewolf on some Linux distribution appears as firefox on windows, which is by far more common) ^[22]
- Installation of an ad-blocker such as uBlock Origin ^[8-1]
- Another good option may be the use of a tool like Adnauseum ^[23], which goes one step further than an ad-blocker and implements more active obfuscation of your data by *actively clicking on advertisements*. However, because of the sheer quantity/throughput it can handle, this has an inverse effect to what advertisers are looking for, as if you appear "interested" in every single advertisement it becomes impossible to narrow down what they should actually show you to effectively advertise to you. In some cases, this may even actively waste advertisers money as, with enough people doing this, the turnover rate of advertising budget to product sold drops exponentially.

There are also a number of lesser settings on *Blackboard* itself you may configure for enhanced privacy:

- Use the most minimal "x" can view my profile on the ^[24] *Blackboard* profile page
- Disable any extra applications you have granted authorisation tokens under the ^[25] tools page, which manages *Blackboard's* own OpenID Connect authentication scopes.

Why take these steps to protect yourself

I am certain that the vast majority of students would feel at the very least, uncomfortable with what *Blackboard* are able to do with their data, or the fact that it remains in their possession

for as long as it does. Aside from the obvious and previously discussed ethical ramifications, this also poses a serious security risk in the event of any form of data breach on *Blackboard's* behalf. For example, if the timetable data of some students were to be released, they could have their precise, predictable location leaked, which I believe many would be uncomfortable with someone acting in bad faith having access to.

Engagement and academic performance data can also evidently be used to predict a given student's future results and may be used to inform employment platforms, such as LinkedIn, about how good of a worker a student may make, which may unfairly reflect their character in an inaccurate way. Personally, I find the potential for academic profiling making it's way to third parties the worst out of *Blackboard's* data harvesting policies.

Conclusion

In conclusion, it should be clear by now that the tracking done by *Blackboard* is incredibly invasive and extensive - and especially so considering that there is no feasible way to opt out of using it for a lot of people. As students, we have no choice but to trust that Blackboard are good people and don't expose our data to anyone malicious, however that becomes incredibly difficult when taking in to account even just the advertising providers that they do mention ^[4-4]. It is nigh impossible to know what might be happening with our data behind the scenes, which presents a serious privacy risk as more and more of our college services are digitised.

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As a potential solution, I would like to implore Trinity to re-think it's use of *Blackboard* as it's primary learning management system in the first place. On top of all the privacy issues raised above, the sheer size of the bundle sent for the web app, along with the generally subpar user experience and implementation of web standards, creates impossible to ignore red flags as someone who is into tech. While I can't find any official figures for how much Trinity spends on *Blackboard* in a given year, one can only imagine it is a significant amount, and seeing as *Blackboard* itself really is not that complex of a concept, I would recommend switching to some form of open source, self hosted alternative to ensure privacy and an excellent user experience for the users.

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1. <https://www.anthology.com/> - Accessed 29th Oct 2025 ↩
 2. https://www.tcd.ie/news_events/articles/trinity-launches-new-online-courses/ - Accessed October 29th 2025 ↩
 3. <https://gilfuseducationgroup.com/history-of-blackboard-and-the-blackboard-learning-system/> - Accessed October 29th 2025 ↩
 4. <https://www.anthology.com/trust-center/privacy-statement> - Accessed October 29th 2025 ↩ ↩ ↩ ↩ ↩
 5. <https://www.tcd.ie/itservices/vle/#accordion2170885> - Accessed 29th October 2025 ↩
 6. *What is the internet doing to me?* Blackboard Page Network Requests filtered to JavaScript files - Accessed 5th November 2025

Status	Method	Domain	File	Initiator	Type	Transferred	Size	Time
200	GET	ultra.content.blackboardcdn.com	6226-ee6135c1b19882.js	script	js	1.43 MB	4...	432 ms
200	GET	ultra.content.blackboardcdn.com	57719-ee6135c1b19882.js	script	js	546.80 kB	1...	431 ms
200	GET	ultra.content.blackboardcdn.com	94700-ee6135c1b19882.js	script	js	233.63 kB	8...	184 ms
200	GET	ultra.content.blackboardcdn.com	app-ee6135c1b19882.js	script	js	1.53 MB	5...	493 ms
200	GET	ultra.content.blackboardcdn.com	initial-load-ee6135c1b19882.js	script	js	654 B	0 B	428 ms
200	GET	ultra.content.blackboardcdn.com	locale-en-ee6135c1b19882.js	outline-8 (script)	js	254.50 kB	1...	172 ms
200	GET	ultra.content.blackboardcdn.com	49358-ee6135c1b19882.js	outline-8 (script)	js	5.83 kB	1...	87 ms
200	GET	ultra.content.blackboardcdn.com	27480-ee6135c1b19882.js	outline-8 (script)	js	3.57 kB	1...	86 ms
200	GET	ultra.content.blackboardcdn.com	66183-ee6135c1b19882.js	outline-8 (script)	js	20.75 kB	7...	87 ms
200	GET	ultra.content.blackboardcdn.com	33049-ee6135c1b19882.js	outline-8 (script)	js	5.80 kB	1...	86 ms
200	GET	ultra.content.blackboardcdn.com	96939-ee6135c1b19882.js	outline-8 (script)	js	19.19 kB	6...	89 ms
200	GET	ultra.content.blackboardcdn.com	95010-ee6135c1b19882.js	outline-8 (script)	js	39.53 kB	7...	76 ms
200	GET	notifi.websockets-eu-central-1.pr...	clientframe.ce39154662094544a3d.js	script	js	60.78 kB	6...	171 ms
200	GET	ultra.content.blackboardcdn.com	64361-ee6135c1b19882.js	outline-8 (script)	js	3.99 kB	9...	15 ms
200	GET	ultra.content.blackboardcdn.com	58140-ee6135c1b19882.js	outline-8 (script)	js	21.14 kB	6...	19 ms
200	GET	ultra.content.blackboardcdn.com	32261-ee6135c1b19882.js	outline-8 (script)	js	3.53 kB	9...	29 ms
200	GET	ultra.content.blackboardcdn.com	35524-ee6135c1b19882.js	outline-8 (script)	js	8.19 kB	2...	28 ms
200	GET	ultra.content.blackboardcdn.com	871-ee6135c1b19882.js	outline-8 (script)	js	7.25 kB	2...	28 ms
200	GET	ultra.content.blackboardcdn.com	7606-ee6135c1b19882.js	outline-8 (script)	js	1.32 kB	1...	25 ms
200	GET	developer.blackboard.com	fileStorage.js	script	js	2.57 kB	1...	91 ms
200	GET	ultra.content.blackboardcdn.com	69292-ee6135c1b19882.js	outline-8 (script)	js	10.26 kB	3...	17 ms
200	GET	ultra.content.blackboardcdn.com	87371-ee6135c1b19882.js	outline-8 (script)	js	4.29 kB	1...	19 ms
200	GET	ultra.content.blackboardcdn.com	1514-ee6135c1b19882.js	outline-8 (script)	js	7.51 kB	2...	25 ms
200	GET	ultra.content.blackboardcdn.com	39769-ee6135c1b19882.js	outline-8 (script)	js	5.96 kB	1...	24 ms
200	GET	ultra.content.blackboardcdn.com	65572-ee6135c1b19882.js	outline-8 (script)	js	10.89 kB	3...	28 ms
200	GET	ultra.content.blackboardcdn.com	72260-ee6135c1b19882.js	outline-8 (script)	js	12.08 kB	4...	28 ms
200	GET	ultra.content.blackboardcdn.com	3774-ee6135c1b19882.js	outline-8 (script)	js	17.58 kB	5...	28 ms
200	GET	ultra.content.blackboardcdn.com	96333-ee6135c1b19882.js	outline-8 (script)	js	5.98 kB	1...	31 ms
200	GET	ultra.content.blackboardcdn.com	46195-ee6135c1b19882.js	outline-8 (script)	js	5.54 kB	1...	31 ms
200	GET	ultra.content.blackboardcdn.com	36182-ee6135c1b19882.js	outline-8 (script)	js	15.58 kB	6...	32 ms
200	GET	ultra.content.blackboardcdn.com	65701-ee6135c1b19882.js	outline-8 (script)	js	3.63 kB	1...	31 ms
200	GET	ultra.content.blackboardcdn.com	locale-en-gb-ee6135c1b19882.js	outline-8 (script)	js	253.53 kB	1...	215 ms
200	GET	learn.content.blackboardcdn.com	18b-jpV=3902.1.0-rel.26+4bdc7a	script	js	1.39 kB	2...	181 ms
200	GET	learn.content.blackboardcdn.com	cookie.js	script	js	1.38 kB	2...	181 ms
200	GET	learn.content.blackboardcdn.com	cdn.js	script	js	2.74 kB	7...	189 ms
200	GET	ltd.blackboard.com	EJALeE2639CF896CFM6371600710B420.jpV=3902.1.0-rel.26+4bdc7a	script	js	248.43 kB	9...	325 ms
200	GET	ltd.blackboard.com	UserDataDWRFeed.jpV=3902.1.0-rel.26+4bdc7a_3902.1.0-rel.26+4bdc7a	script	js	1.41 kB	1...	74 ms
200	GET	ltd.blackboard.com	MarshUpDWRFeed.jpV=3902.1.0-rel.26+4bdc7a_3902.1.0-rel.26+4bdc7a	script	js	1.35 kB	7...	68 ms
200	GET	learn.content.blackboardcdn.com	lttsentagatle.jpV=3902.1.0-rel.26+4bdc7a	script	js	875 B	2...	186 ms
200	GET	learn.content.blackboardcdn.com	engine.jpV=3902.1.0-rel.26+4bdc7a	script	js	12.30 kB	4...	189 ms



7. <https://librewolf.net/> - Accessed 5th November 2025 ↩ ↩
8. <https://ublockorigin.com/> - Accessed 5th November 2025 ↩ ↩
9. <https://nixos.org/> - Accessed 5th November 2025 ↩
10. <https://webpack.js.org/> - Accessed 5th November 2025 ↩
11. `cookie.js` file, which handle's blackboard's cookie implementation - Accessed 5th November 2025

cookie.js

```

/**
 * Sets a Cookie with the given name and value.
 *
 * * name      Name of the cookie
 * * value     Value of the cookie
 * * [expires] Expiration date of the cookie (default: end of current sess
 * * [path]    Path where the cookie is valid (default: path of calling do
 * * [domain]  Domain where the cookie is valid
 * *           (default: domain of calling document)
 * * [secure]  Boolean value indicating if the cookie transmission require
 * *           secure transmission
 */
function setCookie(name, value, expires, path, domain, secure)
{
    document.cookie=name + "=" + escape(value) +
        ((expires) ? "; expires=" + expires.toGMTString() : "") +
        ((path) ? "; path=" + path : "; path=/") +
        ((domain) ? "; domain=" + domain : "") +
        ((secure) ? "; secure" : "");
}

function setRootCookie(name, value, expires, path, domain, secure)
{
    document.cookie= name + "=" + escape(value) +

```

```

        ((expires) ? "; expires=" + expires.toGMTString() : "") +
        "; path=/" +
        ((domain) ? "; domain=" + domain : "") +
        ((secure) ? "; secure" : "");
    }

/**
 * Gets the value of the specified cookie.
 *
 * name Name of the desired cookie.
 *
 * Returns a string containing value of specified cookie,
 * or null if cookie does not exist.
 */
function getCookie(name)
{
    var dc = document.cookie;
    var prefix = name + "=";
    var begin = dc.indexOf("; " + prefix);
    if (begin == -1)
    {
        begin = dc.indexOf(prefix);
        if (begin != 0)
        {
            return null;
        }
    }
    else
    {
        begin += 2;
    }
    var end = document.cookie.indexOf(";", begin);
    if (end == -1)
    {
        end = dc.length;
    }
    return unescape(dc.substring(begin + prefix.length, end));
}

/**
 * Deletes the specified cookie. Will only perform the deletion if
 *
 * a) The cookie exists in the current path; or
 * b) The alwaysDelete flag is specified
 *
 * name name of the cookie
 * [path] path of the cookie (must be same as path used to create
 * [domain] domain of the cookie (must be same as domain used to cr
 * [alwaysDelete] if true, delete the cookie whether it exists in the cur
 */

```

```
function deleteCookie(name, path, domain, alwaysDelete)
{
    if (getCookie(name) || alwaysDelete)
    {
        document.cookie = name + "=" +
            ((path) ? "; path=" + path : "") +
            ((domain) ? "; domain=" + domain : "") +
            "; expires=" + new Date(1).toGMTString();
    }
}
```

12. Pendo.js - from <https://pendo.io> - Accessed 5th November 2025 ↩

GET	ultra-content.blackboardcdn.com	initial-load-e2e135c1b13082.js	script		
GET	cdn.pendo.io	pendo.js	outline_2 (script)	Tracking	
GET	tcd.blackboard.com	favicon.ico	img		

13. <https://www.pendo.io/about/> - Accessed 5th November 2025 ↩
14. <https://support.pendo.io/hc/en-us/articles/360031862272-Install-Pendo-on-a-single-page-web-application> - Accessed 17th November 2025 ↩
15. <https://support.pendo.io/hc/en-us/articles/360032294291-Configure-Track-Events> - Accessed 17th November 2025 ↩
16. js-agent.newrelic.com - Accessed 5th November 2025

GET	js-agent.newrelic.com	nr-spa-1.285.0.min.js	outline_8 (script)	Blocked By uBlock...	
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17. <https://newrelic.com/> - Accessed November 5th 2025 ↩
18. <https://docs.newrelic.com/docs/browser/browser-monitoring/installation/install-browser-monitoring-agent/> - Accessed 17th November 2025 ↩
19. <https://docs.newrelic.com/docs/browser/browser-monitoring/browser-pro-features/session-replay/get-started/> - Accessed 17th November 2025 ↩
20. <https://brainfucksec.github.io/firefox-hardening-guide> - Accessed November 5th 2025 ↩
21. <https://librewolf.net/docs/features/?pubDate=20250819#privacy> - Accessed 5th November 2025 ↩
22. <https://seon.io/resources/3-examples-of-browser-spoofing-and-how-to-detect-them/> - Accessed 5th November 2025 ↩
23. <https://adnauseam.io/> ↩
24. <https://tcd.blackboard.com/ultra/profile> - Accessed 17th November 2025 ↩
25. <https://tcd.blackboard.com/ultra/tools> - Accessed 17th November 2025 ↩